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The paradox between the environmental appeal of bio-based plastic packaging for consumers and their disposal behaviour

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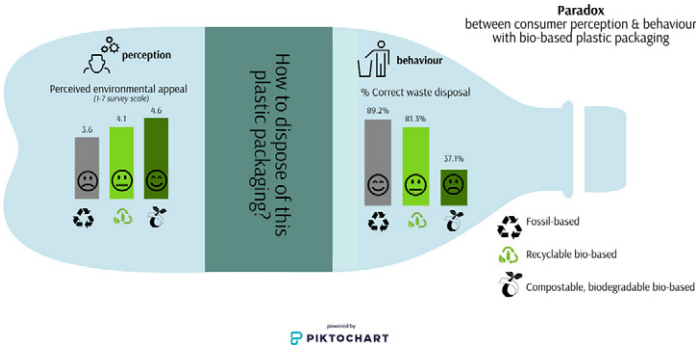
HIGHLIGHTS

- Compostable bio-based packages have the most environmental appeal for consumers.
- Recyclable bio-based and fossil-based packages have similar correct disposal rates.
- Lower correct disposal rates for compostable bio-based packages than fossil-based
- Environmental appeal of packaging does not translate in more correct disposal.
- Greater bio-based familiarity is positively related to adequate consumer disposal.

GRAPHICAL ABSTRACT



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ABSTRACT

To realize the potential environmental benefits that recycling and/or composting bio-based plastic packages can deliver, it is important that consumers view bio-based packaging as environmentally-friendly, but also correctly dispose of the packaging. The current experimental lab-in-the-field study was conducted among German consumers ($n = 281$) and explores whether consumers' perceived environmental benefits of recyclable and compostable bio-based plastic packages match with how consumers dispose of these packages. The results show that consumers only perceive compostable bio-based packages to have more environmental benefits than fossil-based packages. However, consumers dispose of compostable bio-based packages in an incorrect manner (not in line with what is communicated on the packaging label) relatively often. Consumers with a stronger familiarity with bio-based products more often correctly dispose of compostable bio-based packages, but not recyclable bio-based packages, relative to fossil-based packages. Thus, although mainly compostable bio-based plastic packages have strong environmental appeal to consumers, paradoxically this does not translate in the proper disposal actions to fully capitalize on the environmental benefits that bio-based packages can actually deliver. Increasing consumers' bio-based product familiarity might be an avenue to increase the levels of sustainable disposal.

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1. Introduction

Every day, consumers buy and use a lot of products, many of which are packaged in plastic. The packaging of a product can significantly contribute to the environmental impact of the product (e.g. Williams and Wikström, 2011), both in terms of packaging production and waste (Fitzpatrick et al., 2012; Herbes et al., 2018). Particularly disposal waste from plastic packaging strongly affects the environment, as eight million tonnes of plastic waste annually enters the ocean (MacArthur, 2017), a substantial part being plastic packaging waste (Geyer et al., 2017). In general, consumers view the environmental impact from plastic production and waste as a major environmental problem and behavioural solutions are needed to tackle this to combat (ocean) plastic pollution (Heidbreder et al., 2019; Pahl et al., 2017). The majority of plastic packages are produced from fossil-based resources (Spierling et al., 2018), resulting in a large carbon footprint (Pasqualino et al., 2011). A potentially more sustainable alternative are plastic packages made from bio-based materials (Philp et al., 2013). While there is not a single definition of what constitutes a bio-based package (Álvarez-Chávez et al., 2012), bio-based packages are always at least partly based on materials derived from biomass (e.g. Brockhaus et al., 2016). The use of bioplastics for plastic packaging is growing by 20–25% per year, indicating that bio-based materials are increasingly viewed as a viable alternative for fossil-based materials (Arikan and Ozsoy, 2015), even more so when bio-based materials are biodegradable (Karan et al., 2019). There is no clear-cut answer to the exact environmental impact of the use of bio-based materials as this depends on many factors (Hottle et al., 2013). However, prospects for the use of bio-based materials to contribute to a cleaner environment are promising (Philp et al., 2013), as the use of bio-based materials generally has lower greenhouse gas emissions than fossil-based materials (e.g. Bos et al., 2016).

There is some variation in how environmentally-friendly consumers perceive bio-based packaged products to be (Sijtsema et al., 2016), one potential reason being differences in specific types of bio-based packaging. One aspect in which bio-based plastic packages can differ is in their end-of-life disposal options; some types of bio-based plastic packages can be recycled by consumers, while others cannot (Álvarez-Chávez et al., 2012). Also, certain types of bio-based plastic packages are made from materials that can be composted (Álvarez-Chávez et al., 2012). Composting refers to the deliberate biodegradation of organic waste, as opposed to this waste being disposed of in the garbage (White and Simpson, 2013). Thus, one type of bio-based plastic packaging is recyclable bio-based plastic packaging, which mirror their fossil-based counterpart in terms of properties and can be recycled in the end-of-life (post-use) phase (Hottle et al., 2013). A different type is compostable bio-based plastic packaging, which is capable of being biodegraded under specific conditions, which are usually only encountered in an industrial compost facility (Kershaw, 2015). The extent to which the packaging is biodegradable might affect consumer perceptions of environmental friendliness, as consumers positively evaluate biodegradability of packaging (Magnier and Crié, 2015), possibly because of its perceived environmental benefits. However, adequate disposal by consumers of such bio-based packages is needed to actually capitalize on environmental benefits related to sustainable waste disposal (Kale et al., 2007). Given that many consumers are quite unfamiliar with bio-based packaging (Sijtsema et al., 2016), it is not straightforward that consumers actually perceive bio-based packages to be environmentally-friendly and that they know how to correctly dispose of these packages.

In this paper, we aim to gain more insight in the environmental appeal of different types of bio-based plastic packages for consumers relative to fossil-based plastic packages, and whether consumers' disposal decisions match accordingly in the sense that they dispose of bio-based plastic packages in a sustainable manner. Also, we explore whether consumers' familiarity (or lack thereof) with bio-based products affects consumers' perception of the environmental appeal of different types of bio-based packages and their disposal decisions. By gaining such insights, a better understanding can be gained which type of bio-based plastic packaging is most appealing to consumers in terms of environmentally-friendliness, and whether consumers' perceived environmental benefits translate into a sustainable disposal of the packaging. Thus, the main aim of this paper is three-fold, namely to uncover: (1) how environmentally-friendly consumers view different types of bio-based plastic packages relative to fossil-based plastic packages, (2) how consumers dispose of these different types of bio-based plastic packages relative to fossil-based plastic packages and (3) the extent to which consumers' familiarity with bio-based products affects consumers' perceived environmental benefits and disposal behaviour of the different types of bio-based plastic packages relative to fossil-based plastic packages.

How consumers perceive packaging characteristics can affect whether consumers dispose of packaging in a sustainable manner by for instance recycling the packaging (Williams et al., 2018). More specifically, Borgman et al. (2018) have proposed that there is a link between packaging features that convey sustainability and whether consumers engage in sustainable disposal of the packaging, by for instance recycling. Therefore, as a starting point it is important to first establish how much environmental benefits consumers perceive bio-based plastic packaging to have, relative to fossil-based plastic packaging. Consumers appear to associate bio-based materials with being environmentally-friendly, although consumers' opinions differ in how much environmental benefits bio-based packages actually have (Sijtsema et al., 2016). Many consumers are unable to distinguish bio-based from fossil-based packages, because of their unfamiliarity with the concept of bio-based materials (Sijtsema et al., 2016) and the often similar appearance of bio-based packaging and its fossil-based counterpart (Van den Oever et al., 2017). Thus, consumers find it difficult to distinguish bio-based packages from fossil-based packages and might therefore not perceive bio-based packages to have additional environmental benefits relative to fossil-based packages.

Recyclability and compostability are key characteristics that make packaging sustainable in the eyes of consumers (Boesen et al., 2019). Nowadays, fossil-based plastic packages are most often recyclable, which is also communicated on the packaging (Boesen et al., 2019), but they are often not biodegradable and compostable (Geyer et al., 2017). Still, compostable bio-based forms of packaging do exist (e.g., Álvarez-Chávez et al., 2012), and are already increasingly being used by companies (Arikan and Ozsoy, 2015). We argue that compostable bio-based plastic packages have a feature in the form of compostability that can distinguish bio-based packages from fossil-based packages for consumers. In contrast, bio-based plastic packages that are only recyclable can still be hard to distinguish from fossil-based plastic packages for consumers, given their similar appearance (Van den Oever et al., 2017), as well as their similar manner of disposal. Therefore, we formulate the following hypothesis:

H1. Consumers perceive compostable bio-based plastic packages as having more environmental benefits compared to fossil-based plastic packages, while recyclable (non-compostable) bio-based plastic packages are not perceived as having more environmental benefits compared to fossil-based plastic packages.

Few studies have focused on how (sustainable) packages are disposed of by consumers (Martinho et al., 2015). To our knowledge, there has not been a study yet that examines how consumers dispose of different types of bio-based plastic packages, which is particularly relevant given consumers' lack of familiarity with bio-based packages (Sijtsema et al., 2016). In other words, do most consumers indeed adequately dispose of recyclable bio-based plastic packaging in the recycling waste and dispose of compostable bio-based plastic packaging in the organic waste to compost the packaging material? In general, consumers' familiarity with composting waste is lower than with recycling waste (White and Simpson, 2013). More specifically for

packaging, consumers' level of familiarity with a symbol on a packaging label that indicates that packaging is compostable is lower than consumers' familiarity with a symbol that indicates that packaging is recyclable (Boesen et al., 2019). For bio-based packaged products this can be even more so the case, as consumers overall show a lack of familiarity with the concept of bio-based (Sijtsema et al., 2016), whereas the large majority of consumers indicates an awareness of recycling fossil-based packages (Buelow et al., 2010).

Paradoxically, consumers' disposal of compostable bio-based plastic packages can occur in incorrect ways relatively more frequent due to consumers' lack of familiarity with the concept of bio-based in general, as well as with composting, which would (partially) void the potential environmental benefits of compostable bio-based plastic packages. Following our previous reasoning that consumers will find recyclable bio-based plastic packages difficult to distinguish from fossil-based plastic packages, we do not expect a difference in the correct disposal rate between these two types of packaging. Hence, the following hypothesis is formulated:

H2. Compostable bio-based plastic packages are disposed of in an incorrect manner by a larger proportion of consumers, compared to fossil-based plastic packages and recyclable (non-compostable) bio-based plastic packages.

Given that H_1 and H_2 are partly rooted on consumers' lack of familiarity with the concept of bio-based, we propose that a distinction can be made between consumers who are unfamiliar with bio-based products and those who are more familiar with bio-based products. If consumers are more familiar with bio-based products, it becomes less likely that they are unable to distinguish bio-based from fossil-based packages (Sijtsema et al., 2016; Van den Oever et al., 2017) and more likely to perceive bio-based packaging as environmentally-friendly, as they can be expected to be more knowledgeable about what bio-based entails. Building on H_1, we formulate the following hypothesis:

H3a. Consumers' level of bio-based product familiarity strengthens the extent to which consumers perceive compostable bio-based plastic packages as having more environmental benefits compared to fossil-based and recyclable (non-compostable) bio-based plastic packages.

Consumers with a higher level of bio-based product familiarity generally have more positive attitudes towards bio-based products (Reinders et al., 2017). Also, consumers who have more positive attitudes towards the environment are more likely to dispose of packaging in a sustainable manner, presumably because they more strongly consider environmental consequences of disposal (Van Birgelen et al., 2009). Consequently, consumers who have a higher level of bio-based product familiarity can be expected to make more sustainable disposal decisions by correctly disposing of bio-based packaging. Hence, we extend H_2 and formulate the following hypothesis:

H3b. Consumers' level of bio-based product familiarity strengthens the level of correct disposal for compostable bio-based packages compared to disposal of fossil-based and recyclable (non-compostable) bio-based plastic packages.

1.1. Current study

For the current study, we selected three types of packaging that differ in terms of: 1) being bio-based or not and 2) being compostable or not. Thus, the focus is on three types of plastic packaging: fossil-based packaging, recyclable (non-compostable) bio-based packaging and compostable bio-based packaging.

Plastic drinking bottles of mineral water were used as a case study for the current study. Currently, per minute one million plastic bottles are bought globally (Laville and Taylor, 2017) and the large majority of plastic packaging are still used only a single time (Sardon and Dove, 2018) and many end up polluting the planet's ocean (MacArthur, 2017). This makes the need for using more sustainable materials for plastic bottles, and a correct disposal thereof, particularly important from an environmental point of view. More information on the design of the packages is provided in the Method section.

2. Method

2.1. Participants

The study sample consisted of 281 participants (49.8% male, 50.2% female), with an average age of M = 41.99 years (SD = 16.25; range 18–84 years). Participants were recruited in a large shopping street in Cologne (Germany) by a market research company. Participants received a monetary reward for participation in the study. For recruitment of the participants, the instruction was given to the market research company that the study sample should be as representative as possible for the German population in terms of gender, age, education level and income level. Quota sampling was used, which implies that when a certain quota was reached (for instance, enough male participants), people with this characteristic would no longer be approached in the shopping street for study participation. This sampling procedure was used to ensure that the study sample would indeed be representative in terms of these socio-demographic characteristics. Data were collected for two weeks in October 2018. For each participant a study duration of 30 min was planned in the building where the study was conducted. This maximum duration of 30 min was chosen to ensure that participants would not overlap and see each other during the experiment, which would potentially affect their behaviour and thus the study findings.

2.2. Design, procedure and measures

Participants were recruited in a large shopping street and those who agreed to participate went to an office building in the same shopping street where the study would be conducted. The study can be characterized as a lab-in-the-field experiment, where participants are recruited in the field and subsequently perform laboratory tasks which are not part of their everyday environment (Viceisza, 2016). A participant was welcomed by the experimenter and explained that during the study he or she would evaluate a new product, namely a 0.5 l bottle of mineral water. The study had a between-subjects experimental design. Participants received one of three types of plastic packaging containing the product, in line with random assignment to one of the three study conditions: (1) fossil-based, (2) recyclable bio-based, or (3) compostable bio-based.[1] The packages were different only in two ways: 1) whether they are bio-based or not as communicated on the packaging label and 2) whether the packaging needs to be recycled or composted, which was also communicated on the label (see Appendix 1 for the packaging labels). The packaging labels read “Packaging recyclable” (Condition 1), “Packaging recyclable and made from bio-based materials” (Condition 2) and “Packaging compostable and made from bio-based materials” (Condition 3). The actual symbols that indicate whether packaging is to be recycled or composted were used on the labels.

The study consisted of several phases. First, the experimenter asked the participant two face-to-face questions, on what first came to mind when looking at the product and then what came to mind specifically with regard to the packaging. Second, the participant was instructed to go to the laptop in the room and complete Part 1 of an online survey.

[1] There was a fourth study condition in which participants received a carton package. This condition was included to study the effects of the appearance (plastic versus carton) of the packaging on consumers' perceived environmental benefits and disposal actions. This condition is beyond the scope of this paper; therefore, this condition is not included in the results of this paper.

This part of the survey contained questions on purchase intention, product and packaging associations (based on Jerzyk, 2016; Lindh et al., 2016; Prakash and Pathak, 2017; Steenis et al., 2017), perceived environmental benefits (based on Koenig-Lewis et al., 2014) and emotions elicited by the product (based on Koenig-Lewis et al., 2014); all items were measured with a 7-point Likert scale. Perceived environmental benefits were used to test H_1 and H_{3a}. The scale consisted of six items, where participants had to complete the sentence "This package and the material that it is made of …": "contributes to reducing the dependence on non-renewable resources", "benefits the planet to a great deal", "helps to reduce environmental problems", "generates less CO_2-emissions during production than conventional packages", "is beneficial for the environment" and "helps to avoid global warming" (Cronbach's $\alpha = 0.98$).

At the end of this part of the survey, participants were instructed to try out a sample of the product by drinking a glass of water. After drinking a sample of the product, the experimenter exchanged the full package for the same type of package, but now empty and asked the participant to imagine that he or she has finished drinking the bottle. An instruction was given to the participant to dispose of the package in one of the waste bins in the building: (1) paper waste, (2) organic waste (for composting waste), (3) recycling waste or (4) residual waste. Each bin had a large label on it which indicated the type of waste that should be disposed in the bin. We counted disposal to be correct, when participants disposed of the plastic bottle as is communicated on the packaging label: disposal of fossil-based and recyclable bio-based plastic packages were counted as correct when disposed of in the recycling waste, and disposal of compostable bio-based plastic packages was counted as correct when disposed of in the organic waste.

After disposal, the participant was asked face-to-face by the experimenter why a particular waste bin was chosen and whether they had incurred any difficulties in this decision-making process. Subsequently, the participant was instructed to return to the laptop and to answer some final questions in the online survey which measured manipulation checks on whether the packaging is made out of bio-based materials and whether the packaging is compostable, and a measure of participants' level of bio-based product familiarity (based on Yoo et al., 2000; Cronbach's $\alpha = 0.69$[2] (3 items)).

The different phases of the study are summarized by means of a flow chart (see Fig. 1).

3. Results

3.1. Manipulation checks

First, we checked whether both types of bio-based packages were indeed viewed as having been made from bio-based materials, more so than the fossil-based package. There was a statistically significant difference between the three types of packages on the degree to which participants perceived the package as having been made out of bio-based materials: $F(2, 278) = 12.71$, $p < .001$. Post-hoc tests (Bonferroni) indicated that participants indeed viewed compostable bio-based packages (M = 4.99) as having been made from bio-based materials to a greater extent than fossil-based packages (M = 3.42), $p < .001$. However, recyclable bio-based packages (M = 4.14) were not viewed as having been made from bio-based materials to a greater extent than fossil-based packages (M = 3.42), $p = .070$.

Second, we checked whether compostable bio-based packages were indeed viewed as being more compostable than the two non-compostable forms of packaging. There was a statistically significant difference between the three types of packages on the extent to which participants perceived the package as being compostable: $F(2, 278) = 23.59$, $p < .001$. Post-hoc tests (Bonferroni) indicated that indeed compostable bio-based packages (M = 5.20) were perceived as being compostable to a greater extent than recyclable bio-based packages (M = 3.31; $p < .001$) and fossil-based packages (M = 3.44; $p < .001$).

3.2. Perceived environmental benefits of packaging

To test H_1, we conducted an Analysis of Variance (ANOVA) which showed a significant difference between the types of packages in how many environmental benefits participants perceived the packaging to have: $F(2,278) = 6.14$, $p = .002$. Post-hoc tests (Bonferroni) showed that as hypothesized (H_1), participants viewed compostable bio-based packages ($M = 4.55$) as having more environmental benefits than fossil-based packages ($M = 3.56$): contrast = 0.99, $p = .002$. There was no significant difference in perceived environmental benefits between recyclable bio-based packages ($M = 4.12$) and fossil-based packages ($M = 3.56$): contrast = 0.56, $p = .154$.[3]

3.3. Waste disposal

Subsequently, we tested the hypothesis that compostable bio-based plastic packages are disposed of in an incorrect manner by a larger proportion of consumers, compared to fossil-based and recyclable, non-compostable plastic packages (H_2). Of the participants who had a compostable bio-based package only 37.1% disposed of the package correctly in the organic waste (54.6% chose recycling waste & 8.2% chose residual waste), significantly less than the 89.2% whom disposed of fossil-based packages correctly in the recycling waste (5.4% chose organic waste & 5.4% chose residual waste): $\chi^2(1) = 55.14$, $p < .001$, and the 81.3% whom correctly disposed of recyclable bio-based packages (11.0% chose organic waste & 7.7% chose residual waste): $\chi^2(1) = 37.79$, $p < .001$, providing support for H_2 (Fig. 2). There was no significant difference between the proportion of participants whom correctly disposed of recyclable bio-based packages (81.3%) and whom correctly disposed of fossil-based packages via disposal in the recycling waste (89.2%): $\chi^2(1) = 2.31$, $p = .129$.

To explore to what extent the type of plastic packaging and disposal are connected via participants' level of perceived environmental benefits of the packaging, we conducted a mediation analysis with disposal as the dependent variable, type of packaging as the multi-categorical independent variable (with D1 = recyclable bio-based relative to fossil-based and D2 = compostable bio-based relative to fossil-based) and perceived environmental benefits as the mediator. The PROCESS macro for SPSS (Hayes, 2012) was used for the analysis (model 4). The analysis showed a significant indirect effect of type of packaging via perceived environmental benefits for both categories of the multi-categorical independent variable (D1: $B = 0.14$, 95% bootstrapped CI [0.004, 0.358], D2: $B = 0.19$, 95% bootstrapped CI [0.024, 0.447]), with a significant positive relation between perceived environmental benefits and likelihood to correctly dispose of the plastic packaging ($B = 0.20$, $p = .016$).

To gain an initial, exploratory understanding of possible reasons behind participants' (in)correct disposal decisions, the experimenter asked why a particular waste bin was chosen and whether participants

[2] The original scale consisted of 4 items, but reliability analyses indicated that one item which was reverse coded ("I have difficulty in imagining bio-based packaged products in my mind") made the scale unreliable (Cronbach's $\alpha = 0.46$). This item was subsequently left out, which resulted in a more reliable scale (Cronbach's $\alpha = 0.69$), consisting of 3 items: "I can recognize bio-based packaged products among other competing products", "I am aware of bio-based packaged products", "Some characteristics of bio-based packaged products come to my mind quickly".

[3] Although it was not the main focus for the study reported in the paper, emotions and product purchase intentions were also measured. There was no significant difference between the three types of packaging regarding positive emotions with respect to the packaging ($F(2, 278) = 2.60$, $p = .076$) or purchase intention ($F(2, 278) = 1.18$, $p = .310$). There was a significant difference regarding negative emotions with respect to the packaging ($F(2, 278) = 3.51$, $p = .031$), with post-hoc tests (Bonferroni) indicating that less negative emotions were experienced regarding the compostable bio-based package (M = 1.80) relative to the fossil-based plastic package (M = 2.31): contrast = −0.51, $p = .034$.

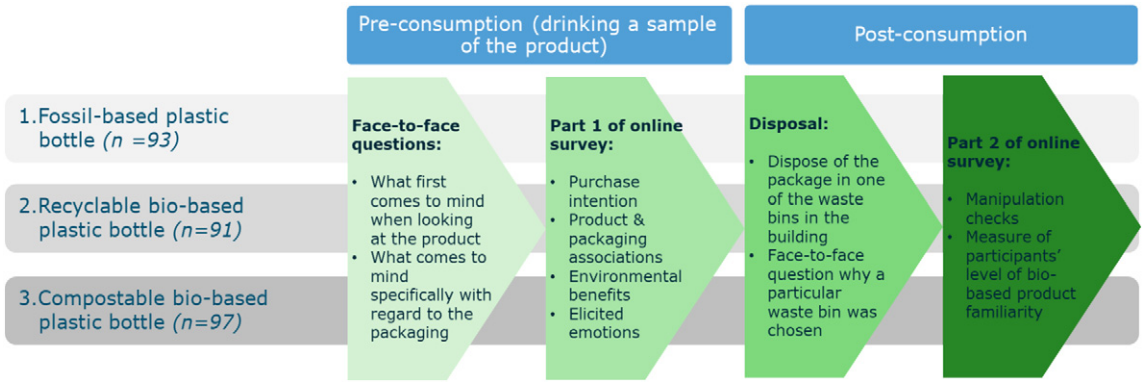

Fig. 1. Flow chart of experimental procedure.

had any difficulties in deciding which waste bin to choose. Given the relatively high level of incorrect disposal of compostable bio-based packages, we particularly focused on possible reasons underlying these incorrect disposal decisions. The quotes reported below were made by a selection of participants (thus not by all participants), and were selected because they give initial, exploratory insights into the decision-making process of disposing the packaging material in a particular waste bin.

Several participants who had a compostable bio-based package provided answers that indicated that they did see that the package was compostable, but they indicated that plastics in general should always go in the recycling waste and therefore did not choose the organic waste bin:

> "Even though it is compostable, I would count it as plastic and therefore chose the recycling waste bin."

> "Because it is a plastic bottle, I would first choose recycling it, but I was unsure how to pay attention to separating waste when packaging is compostable."

> "Because the bottle is made of plastic, I would throw it in the recycling waste. But since it would be compostable, I would actually have to have thrown it in the organic waste."

Furthermore, some participants provided answers that indicated they could not imagine that a plastic bottle can be composted, which also might have contributed to the high level of incorrect disposal of compostable bio-based plastic packages:

> "Because the bottle is made of plastic and plastic belongs in the recycling waste. I do the same at home. At first I had difficulty, because I read the note on the bottle that it is made of compostable materials. First thought to put it in the organic waste, but I could not imagine that with plastic."

> "I was unsure, it is a plastic bottle which said the packaging is compostable, but I find it hard to imagine that."

3.4. Bio-based product familiarity

To test H_{3a}, a moderation analyses with a multi-categorical independent variable was performed, with the fossil-based package condition as the reference group and using 5000 bootstrap samples, following the procedure of Hayes and Preacher (2014). The PROCESS macro for SPSS (Hayes, 2012) was used for the analysis. Following Aiken et al. (1991), we mean-centered the moderator variable bio-based product familiarity. Two dummies were used for the analysis, with D1 = recyclable bio-based relative to fossil-based and D2 = compostable bio-based relative to fossil-based. The results are reported in Table 1. As can be seen in Table 1, there was no significant interaction effect between bio-based product familiarity and D2 (compostable bio-based relative to fossil-

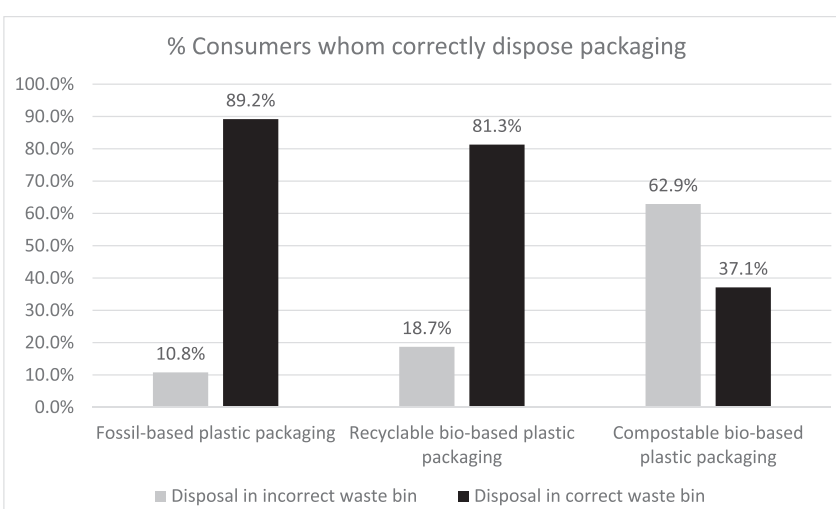

Fig. 2. Consumer disposal of packaging.

Table 1
Results of moderation analysis bio-based product familiarity on perceived environmental benefits.

Variable	B	SE	Sig.
D1 (bio-based recyclable relative to fossil-based)	0.56	0.29	0.051
D2 (bio-based compostable relative to fossil-based)	0.99	0.28	0.001
Bio-based product familiarity	0.11	0.27	0.452
D1 ∗ Bio-based product familiarity	−0.19	0.22	0.377
D2 ∗ Bio-based product familiarity	0.18	0.20	0.368

Table 2
Results of moderation analysis bio-based product familiarity on correct disposal of packaging.

Variable	B	SE	Sig.
D1 (bio-based recyclable relative to fossil-based)	−0.65	0.46	0.166
D2 (bio-based compostable relative to fossil-based)	−2.79	0.42	0.001
Bio-based product familiarity	−0.29	0.27	0.286
D1 ∗ Bio-based product familiarity	0.60	0.35	0.087
D2 ∗ Bio-based product familiarity	0.65	0.32	0.041

based) on perceived environmental benefits of the plastic packaging, which is not in line with H_{3a}.

Similar analyses were conducted to test $H_{3b,}$ though now disposal behaviour was included as the dependent variable. As can be seen in Table 2, the interaction between D2 (compostable bio-based relative to fossil-based) and bio-based product familiarity was significant: $B = 0.65$, $p = .041$. We employed a simple slopes analysis to interpret this interaction. Participants with a relatively high level of bio-based product familiarity (one standard deviation above the mean) were more likely to correctly dispose of compostable bio-based packages ($B = -0.92$, $p = .001$), compared to participants with a relatively low level of bio-based product familiarity (one standard deviation below the mean): $B = -1.82$, $p < .001$, in line with H_{3b} (Fig. 3).[4]

4. Discussion

The use of bio-based materials to produce plastic packaging can be beneficial for the environment, as it for instance contributes to lower greenhouse gas emissions relative to the use of fossil-based materials (Bos et al., 2016; Philp et al., 2013). This is particularly relevant, given significant environmental problems as a result of plastic production and waste (Heidbreder et al., 2019; MacArthur, 2017). To fully capitalize on the potential environmental benefits that bio-based plastic packages can deliver, it is important that consumers do not only perceive these packages as being environmentally-friendly, but also dispose of the packages correctly (i.e. in line with what is communicated on the packaging label) by recycling or composting these packages. The current study aimed to add to the literature by gaining a better understanding of how consumers perceive different types of bio-based plastic packaging in terms of its environmental appeal relative to fossil-based plastic packaging, and whether consumers also dispose of bio-based packaging in line with what is communicated on the packaging label and (depending on the type of package) recycle or compost it. Furthermore, we also examined the role of consumers' familiarity with bio-based products (or lack thereof) in how they view different types of bio-based plastic packaging and dispose of them.

Previous studies have shown that consumers generally perceive more environmental benefits from packaging made from bio-based materials as compared to packages that are made from fossil-based plastics (Sijtsema et al., 2016). The current study extends these findings and shows that only compostable bio-based packages, but not recyclable (non-compostable) bio-based packages, are perceived by consumers to have more environmental benefits compared to fossil-based packages. This finding is in line with Magnier and Crié (2015), who found that consumers have favourable views towards biodegradability of packaging, and in line with Orset et al. (2017) who found that consumers are willing to pay a price premium for biodegradable plastic packaging. More importantly, this finding aligns with Herbes et al. (2018) who found in a cross-country comparative study that materials that are renewable as well as biodegradable are rated highly by consumers, while materials made from renewable but not biodegradable material received slightly lower scores. Taken together, these findings suggest that compostability can be a distinguishing feature to set bio-based plastic packages apart from its fossil-based counterpart and provide an opportunity to make the potential environmental benefits of bio-based plastic packages more salient for consumers. Recycling a bio-based package appears to not be a distinguishing feature in itself, as most fossil-based plastic packages are also recyclable (Boesen et al., 2019); our manipulation check corroborated this as in the current study recyclable (non-compostable) bio-based plastic packages were not viewed by consumers as having been made from bio-based materials to a greater extent than its fossil-based counterpart. Thus, making plastic packages from bio-based materials that are compostable and biodegradable contributes to making bio-based plastic packages more environmentally-friendly in the eyes of consumers, relative to fossil-based plastic packages.

Paradoxically, although compostable bio-based plastic packages are perceived by consumers as having relatively sizable environmental benefits, consumers' disposal behaviour of this type of package does not match accordingly. In the current study, the proportion of consumers that correctly disposed of compostable bio-based packages is significantly less compared to the proportions of consumers that correctly dispose of fossil-based packages and recyclable bio-based packages. This can be a result of consumers already being quite familiar with recycling plastic packages (Boesen et al., 2019; White and Simpson, 2013), and consumers not being able to distinguish between bio-based and fossil-based plastic packages when they are both recyclable (Van den Oever et al., 2017). However, the combination of plastic packages being both bio-based as well as compostable appears to lead to a relatively strong ambivalence among consumers regarding what to do in terms of waste disposal. This is also reflected in our exploratory findings based on the qualitative answers of consumers regarding their reasons for their disposal decisions. These findings show that some consumers did see that the plastic packaging is compostable, but believed that plastic should always be recycled (and not composted). Additionally, some consumers could not imagine that plastic packages could be compostable and biodegradable in the first place and therefore chose to not dispose of the compostable bio-based package in the organic waste bin. This suggests that it is needed to inform and convince consumers that certain forms of plastic have the potential to be composted, in order for them to dispose of compostable bio-based plastic packages in the organic waste.

Taken together, our findings indicate a paradoxical situation. Features of compostability of bio-based plastic packages result in stronger consumer perceptions of environmental benefits compared to fossil-based plastic packages, but disposal behaviour goes wrong more often compared to the disposal of recyclable bio-based plastic packages. This paradox is amplified even further by the (exploratory) finding that even though there is a positive association between consumer perceptions of environmental benefits of the packaging and the likelihood that the packaging is correctly disposed of, compostable bio-based plastic packages are disposed of incorrectly most often, despite consumers perceiving this type of packaging to have the most environmental benefits. In the literature, a link between sustainable packaging

[4] To check for robustness of the findings, the Area Under the Curve (AUC) was calculated in Receiver-Operator Characteristic (ROC) analysis in SPSS. To calculate the AUC, predicted probabilities for disposal were used as the test variable and actual correct or incorrect disposal as the state variable. The predicted probabilities were generated from binary logistic regression analysis with disposal as the dependent variable, and perceived environmental benefits, bio-based product familiarity, D1 and D2 as covariates. This ROC analysis generated an AUC of 0.841 for compostable bio-based relative to fossil-based (D2) and an AUC of 0.609 for recyclable bio-based relative to fossil-based (D1).

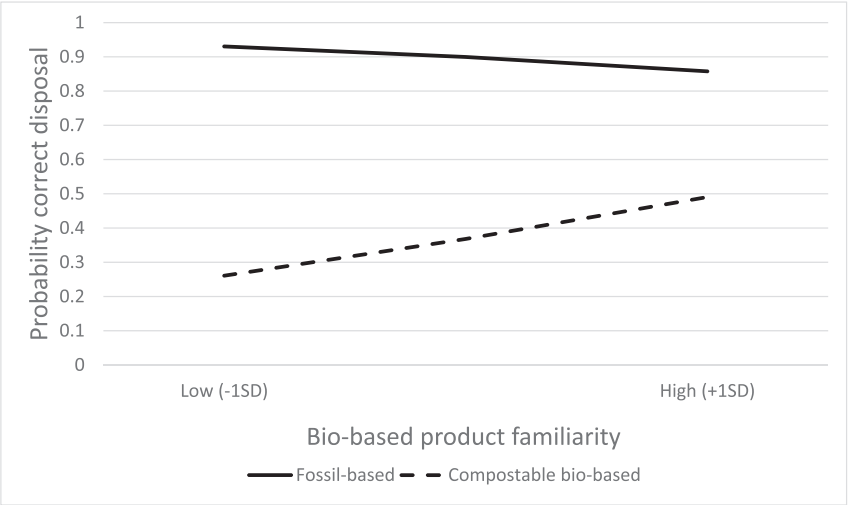

Fig. 3. Effect of packaging type and level of bio-based product familiarity on probability of correct disposal.

characteristics and sustainable behavioural intentions has been previously established (e.g. Konuk et al., 2015; Reinders et al., 2017), similar to the positive association between perceived environmental appeal and correct waste disposal in the current study. However, the current study suggests that the level of perceived environmental benefits for compostable bio-based plastic packaging is not sufficiently high (yet) in order to compensate for other possible reasons that lead consumers to often incorrectly dispose of this specific type of packaging, thus leading to a relatively low correct disposal rate for compostable bio-based plastic packages. In this respect, the current study shows the relevance of consumers' familiarity with bio-based products in understanding why consumers dispose of compostable bio-based packages more often incorrectly than recyclable (bio-based and fossil-based) packages. The higher consumers' level of bio-based product familiarity is, the higher the probability is that compostable bio-based packages are correctly disposed of, relative to the correct disposal rate of fossil-based packaging. However, as is shown by recent previous studies, familiarity with bio-based products and packaging is generally low (Herbes et al., 2018; Sijtsema et al., 2016). In addition, familiarity with the different symbols or logos on packaging could also play a role. Boesen et al. (2019) showed that consumers are more familiar with the recycling symbol on packaging, but less with the compostability symbol; these symbols were also used for the current study. This indicates a relevance to increase consumers' familiarity with the compostability symbol and/or to make the symbol more specific in terms of how consumers can compose the packaging.

Finally, consumers' level of bio-based product familiarity does not strengthen the extent to which they perceive compostable bio-based forms of plastic packaging as having more environmental benefits, relative to fossil-based plastic packaging. A potential explanation might be that consumers who are (very) familiar with bio-based products, are also more familiar with the fact that the degree of environmental benefits of bio-based materials depends on many factors (Hottle et al., 2013), leading them to not perceive compostable bio-based plastic packages as more environmentally-friendly without information on for instance how long it takes for the package to decompose.

4.1. Limitations and future research

The current study brought forward a number of recommendations for future research, and has some limitations that can also be addressed in future research. The findings of the current study reveal that compostable bio-based plastic packaging is perceived as more environmentally-friendly compared to recyclable bio-based plastic packaging, though they are disposed of incorrectly (i.e. not in line with what is communicated on the packaging label) more often. It is possible that consumers' knowledge gap regarding compostability differs across countries due to for example regulations, and familiarity with the concept of composing waste. Specifically, many countries (or municipalities within certain countries) currently do not already have the necessary industrial composting facilities and/or separate waste collection methods (Rujnić-Sokele and Pilipović, 2017). Consequently, no generic procedure of how to dispose of compostable packaging currently exists for consumers. Cross-cultural research is therefore necessary to explore whether the findings are generalizable across different countries. Moreover, while we did measure consumers' level of bio-based product familiarity, we did not measure their familiarity with compostability as a waste disposal method. Future research can test whether familiarity with compostability affects the level of correct disposal rate for compostable bio-based plastic packages. This is particularly relevant given that many consumers indicated that they could not imagine that plastic bottles could be compostable, hence leading to an incorrect disposal of compostable bio-based plastic packages. For such consumers, unfamiliarity with bio-based as well as compostability might have affected their disposal behaviour.

In the current study waste bins were placed in the building where the study was conducted. However, waste bins are not always present in the direct vicinity of consumers when they want to dispose of plastic packaging. In general, infrastructure for composting bioplastics is not yet widely available (Álvarez-Chávez et al., 2012); in many countries organic waste bins on the streets are (far) less common than recycling and residual waste bins. However, given that separate collection of organic waste will become mandatory in the European Union by 2023, it is expected that the necessary infrastructure will be further developed in the upcoming years and become more widespread in Europe (Sand, 2019). Currently, compostable packaging are more likely to end up being a sustainable alternative for non-compostable, fossil-based packaging in closed-loop systems (Sand, 2019), such as an office building which was used for the current study. In such closed-loop systems, separate waste disposal and sorting of compostable packaging is easier to manage than in open-loop systems, while minimizing contamination of organic waste, which is critical for effective composting, is also easier to realize in closed-loop systems (Sand, 2019). Hence, compostable packaging is typically more likely to end up in necessary industrial

compost facilities in such closed-loop systems. While home composting can theoretically also be suitable to compost this type of waste, the conditions to support composting waste at home are typically less compared to industrial compost facilities, thus home composting is typically not advisable for compostable, biodegradable packaging (Sand, 2019). Such contextual factors can be further explored in future research to test what consumers do in terms of waste disposal in different types of contexts.

Furthermore, in the current study plastic bottles were used. Although plastic bottles exemplify a highly relevant environmental issue which is recognised by many (e.g. Laville and Taylor, 2017), future research is necessary to explore whether the findings of the current study hold for different types of packaging. Many consumers are familiar with recycling plastic bottles, which could also be seen in consumers' responses to the question why they chose a certain waste bin. But for other types of packaging, such as single-use packaging like packaging of candy bars, meat and fresh produce, consumers are most likely less familiar with recycling the packaging. For these types of packaging, consumers might not recycle recyclable bio-based packaging to the same extent as recyclable bio-based plastic bottles. Additionally, in some countries a deposit system for plastic bottles exists (which was also the case in the current study), while in other countries such a deposit system does not exist. This can also affect how consumers dispose of this specific type of plastic packaging in terms of disposal in the recycling and/or organic waste, since the presence of a deposit system can affect waste disposal rates.

Future research can further explore the underlying mechanisms of the consumer perceptions of different types of bio-based packages that were revealed in the current study. For example, does making the environmental benefits of bio-based packaging more specific, for instance on the packaging itself, lead to higher perceived environmental benefits among consumers (construal level theory; Chang et al., 2015)? Furthermore, future research can further disentangle how the different elements of bio-based plastic packaging affect the extent to which consumers perceive the packaging to have environmental benefits, by designing a choice-experiment in which different types of environmental benefits are stressed on the packaging. Different types of environmental benefits include the percentage of bio-based materials used, the disposal method, but also the environmental impact in terms of CO_2-emissions. Such type of future research can also give more insights into why consumers perceive compostable bio-based packaging to have the most environmental benefits in the current study, relative to the other types of packaging.

The current study focused on grasping perceptions and measures current behaviour in terms of waste disposal, though future studies might focus on finding ways to change behaviour and increase the rate at which consumers correctly dispose of bio-based plastic packages. For example, behavioural experiments may help to obtain insight in what types of communication on the packaging label, design of the recycling and composting symbol on the label, or types of nudges in an environment where recycling and organic waste bins are present, can help consumers to correctly dispose of (compostable) bio-based packages. Finding ways to increase the correct disposal rate of compostable bio-based packages is important, to prevent that part of the environmental benefits that these packages can actually have will be voided.

Sensitivity analysis was conducted, through calculation of the AUC, to explore the robustness of the findings. According to Streiner and Cairney (2007), an AUC between 0.5 is 0.7 is considered low, while an AUC between 0.7 and 0.9 is moderate, and an AUC of 0.9 or higher is considered high. The AUC that we calculated regarding waste disposal for compostable bio-based packaging relative to fossil-based packaging was moderate (0.841), but the AUC was relatively low (0.609) regarding waste disposal for recyclable bio-based packaging relative to fossil-based packaging. Thus, particularly the findings regarding consumer disposal for recyclable bio-based packaging relative to fossil-based packaging should be interpreted with caution. In the current study, the level of correct disposal by consumers did not significantly differ between recyclable bio-based packaging and fossil-based packaging, but given the relatively low AUC regarding waste disposal for recyclable bio-based packaging relative to fossil-based packaging, it is important to replicate this particular finding in future research.

Finally, although the instruction was given to the market research company that collected the data to ensure that the study sample is representative in terms of socio-demographic characteristics, the sample size was relatively small. Future studies would benefit from first conducting a power analysis, to determine an appropriate sample size.

4.2. Conclusions

Consumer perceptions of environmental benefits are highly relevant for consumer acceptance. The current study shows that not all bio-based plastic packages are perceived as being more environmentally-friendly than their fossil-based counterpart. Making the packaging of bio-based materials that are compostable, strengthens consumer perceptions of the environmental benefits of the packaging. Alternatively, consumers do not perceive plastic packaging that is only recyclable (but not compostable) to have greater environmental benefits relative to fossil-based plastic packages. Thus, practitioners can use compostability as a feature to strengthen the environmental appeal of bio-based plastic packaging. Paradoxically, the actual environmental benefits of such compostable bio-based plastic packages might be partly voided, because consumers relatively often dispose of them in the wrong waste bin, whereas the level of adequate disposal is similar for recyclable bio-based plastic packages and fossil-based plastic packages. Thus, in terms of disposal, steps are still needed to encourage consumers to dispose of compostable, biodegradable plastic packages in the appropriate waste bin so that these packages end up in industrial compost facilities where the conditions to support composting waste are maximized. Part of the solution might lie in increasing familiarity with bio-based products, as consumers who are more familiar with the concept of bio-based did correctly dispose of the packages more often. One instrument that can be used to increase this familiarity is labelling on packaging. However, as consumers' knowledge on packaging disposal labels turns out to be limited and currently different label schemes exist in different countries (see Boesen et al., 2019), all involved actors (i.e., producers, retailers and policy makers) are asked to take action to develop communication tools that provide consumers with the knowledge needed to make informed disposal choices.

Bio-based materials have great potential benefits for the environment, also in the eyes of the consumer. However, to capitalize on these environmental benefits future steps are still needed to increase consumers' familiarity with compostable bio-based products and have consumers correctly dispose of these packages, to make it more likely that these types of plastic packages will indeed end up being a sustainable alternative.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. Packaging labels corresponding with the study conditions

Condition 1. Fossil-based plastic package.

Condition 2. Bio-based recyclable plastic package.

Condition 3. Bio-based compostable plastic package

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